H.W.C.M

System definition

The Human-wildlife conflict management system is software designed to help manage and mitigate conflicts between humans and wildlife. These conflicts can arise in various situations, such as when wildlife species come into contact with human settlements, infrastructure, or agricultural areas.

Note:Test conditions are high-level descriptions of what is being tested and why. They describe the specific conditions or situations that need to be tested, without going into the details of how the testing will be performed. Test conditions are usually defined during the test planning phase and help ensure that all important aspects of the system are tested.

Test cases are detailed instructions for performing a specific test. They describe the steps that need to be taken, the expected results, and any other relevant information needed to perform the test. Test cases are developed based on the test conditions and are used during the test execution phase to verify that the system functions as expected.

Objectives

Prevention/mitigation

- The system must prevent conflicts between humans and wildlife by:
    - Informing users of animal movements
    - Notifying users of the causes of animal movements and how they can prevent it
        - Deforestation
        - Overuse of water sources
        - overhunting

Damage control

- The system must provide damage control measures such as:

    - alert users in the area of recent conflicts
    - Inform users how to respond to various conflicts, e.g:
        - How to respond if hyenas are attacking

Tracking and observation

- The system must be able to track and observe animal movements by:
    - Identifying the animal's current territories
    - Track when animals are expanding beyond their expected territory
    - Provide observation methods to users
    - The system should produce charts and analytics of conflicts
    - Manage conflict details e.g:
        - Track the number of livestock deaths
        - Track the number of human attacks
    - Track the history of conflicts

Requirements

Title of Requirement	Description	Target users	Priority	Priority Level
1. Login/Signup	Users must be able to create an account and log in to the system in order to access its features and functionality.	government officials, and local residents.	High	Functional
2. Wildlife database	The system must include a comprehensive database of wildlife species that are known to cause conflicts with humans, including information on their behavior, ecology, and management strategies.	government officials, and local residents.	High	Functional

3. Prevention strategies	The system must include a range of prevention strategies, such as fencing, lighting, and the removal of attractants and is tailored to local conditions.	Local residents and government officials	High	Fuctional
4. Mitigation strategies	The system must include a range of mitigation strategies, such as relocation, hazing, and habitat modification and that can be tailored to local conditions.	Local residents, and government officials.	High	Function al
5. Intervention strategies	The system must include a range of intervention strategies, such as trapping, hunting, and lethal removal and can be tailored to local conditions	government officials	High	Function al
6. Feedback mechanism	The system must include a mechanism for soliciting feedback from local residents, wildlife experts, and other stakeholders, in order to identify areas for improvement and ensure that the system is meeting their needs.	Government officials and local residents	Meduim	Collabor ative
7. Cultural sensitivity	The system must be designed to be culturally sensitive, taking into account the social and cultural context of the communities in which it is being implemented.	Local residents and government officials.	High	Social
8. Analytics	The system must include tools for analyzing damage done, identify problem animals and nature of attack.	Local residents and government officials.	Meduim	Analytica l

End users

- Himba people(Local residents): the primary populous that lives in the system intended use environment.
- Lodge populous: those working and staying at the lodge in the system intended use environment

Environment definition

Terrain

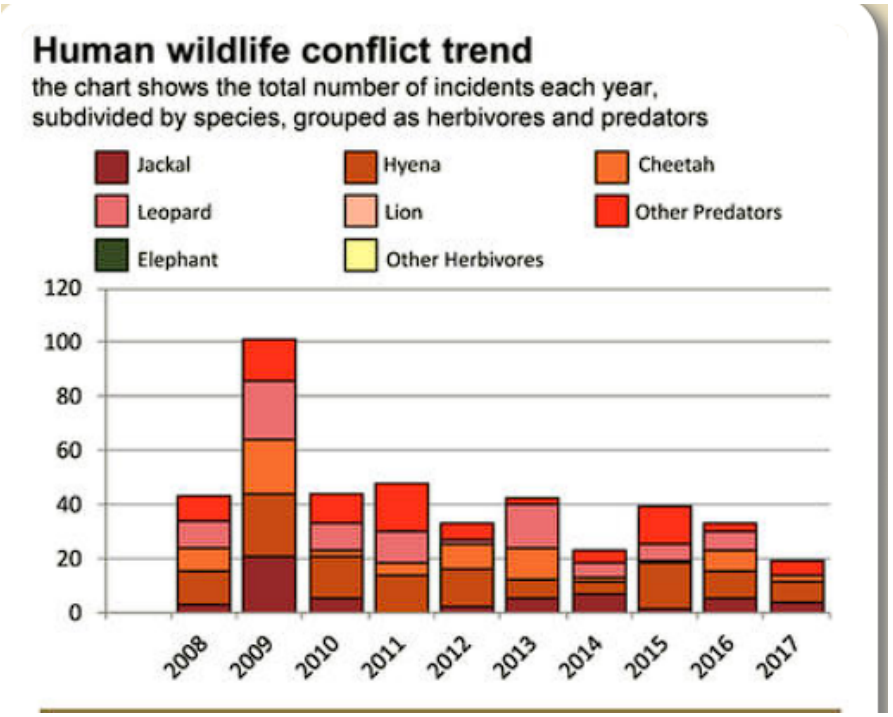
Marienfluss lies at the extreme northwest of the Kunene region. The Marienfluss is a large valley that provides grazing for the cattle and goats of the indigenous Himba population. The Conservancy was registered in 2001 and covers 3,300 km2. There are a mere 340 people in the area – around one person per square kilometer. Grasslands dominate the valleys, with dunes covering the western section. Income is derived from conservation hunting and tourism.

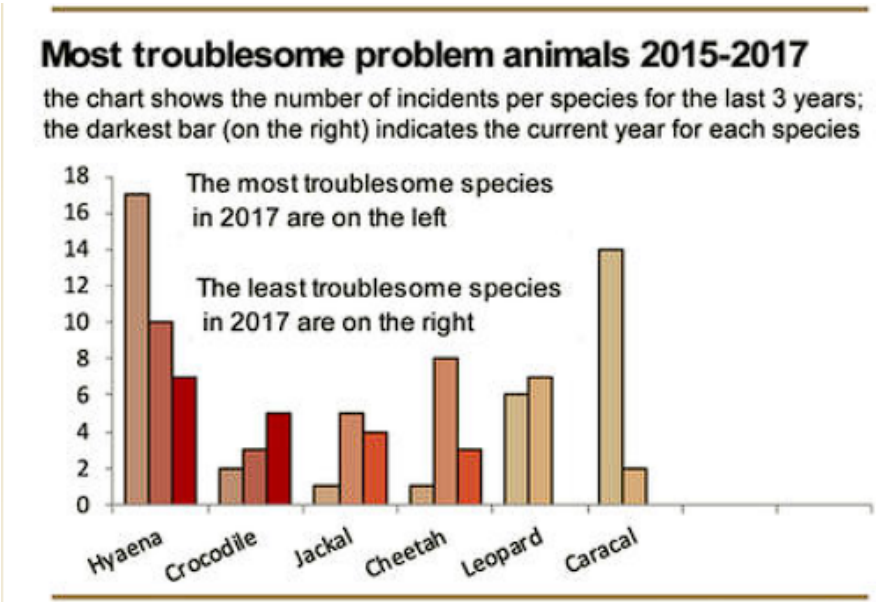
Climate and weather

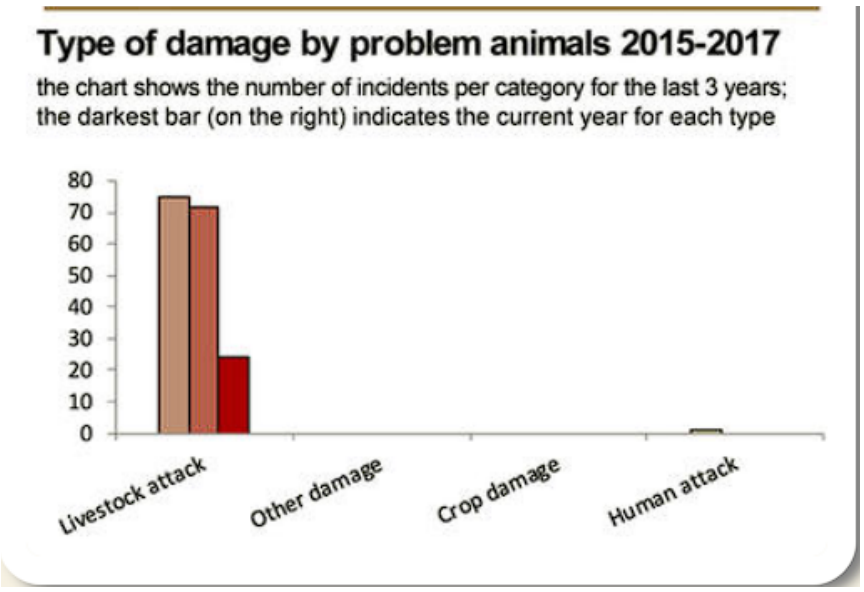
Kunene is one of the warmest regions in Namibia with an average daily high temperature of 33 degrees. It is a warm or hot all year long.

Network

Wildlife


Human wildlife conflict trend
the chart shows the total number of incidents each year, subdivided by species, grouped as herbivores and predators
Jackal
Hyena
Cheetah
Leopard
Lion
Other Predators
Elephant
Other Herbivores
120
100
80
60
40
20
0
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017


Most troublesome problem animals 2015-2017
the chart shows the number of incidents per species for the last 3 years; the darkest bar (on the right) indicates the current year for each species
The most troublesome species in 2017 are on the left
The least troublesome species in 2017 are on the right
18
16
14
12
10
8
6
4
2
0
Hyaena
Crocodile
Jackal
Cheetah
Leopard
Caracal



Test objectives

- Evaluate the accuracy of wildlife density estimates in grassland environments to ensure that the system is providing reliable information for management decisions.

- Assess the performance of the system in identifying and reporting wildlife conflicts in real-time to ensure that the system can provide timely alerts to management staff.

- Test the user interface and user experience of the system to ensure that it is intuitive and easy to use for park rangers or other staff who may not have extensive technical knowledge.

- Evaluate the performance of the system in extreme weather conditions, such as high temperatures or strong winds, to ensure that it remains reliable and functional under all operating conditions.

Variables

1. Eniromental factors:
    - Grasslands dominate the valleys, with dunes covering the western section.
    - Kunene is one of the warmest regions in Namibia with an average daily high temperature of 33 degrees. It is a warm or hot yearlong.
2. User behavior(Local residents):
    - They may not know how to use system correctly leading to errors.
3. Data quality(All research was done on the internet):
    - If data is inaccurate or incomplete, the system may not be able to accurately predict or detect human–wildlife conflicts.

Test Conditions

1. Temperature tolerance testing:
Test the system's ability to function effectively in high-temperature environments, including stress testing of electronic equipment and software under high-temperature conditions.

2. Grassland habitat modeling:
Test the accuracy and reliability of the system's habitat modeling tools for grassland environments, including accuracy of wildlife density estimates and behavior predictions.

3. Dune habitat modeling:
Test the accuracy and reliability of the system's habitat modeling tools for dune environments, including accuracy of wildlife density estimates and behavior predictions.

4. Wildlife behavior analysis:
Test the system's ability to accurately analyze and predict the behavior of key wildlife species, including their movements, feeding patterns, and interactions with human activity.

5. Conflict detection:
Test the system's ability to detect and alert users to potential wildlife-human conflicts, including both high-risk and low-risk situations.

6. Prevention and mitigation testing:
Test the effectiveness of the system's prevention and mitigation strategies for reducing wildlife-human conflicts, including testing of fencing, lighting, and other deterrent measures.

7. Intervention testing:
Test the effectiveness of the system's intervention strategies for managing wildlife-human conflicts, including testing of trapping, relocation, and lethal removal.

8. Data management and reporting:
Test the system's ability to collect, manage, and report on data related to wildlife-human conflicts, including accuracy of data collection, data management tools, and reporting features.

9. User testing: //(Not sure if it should be included)
Test the usability and functionality of the system from the perspective of various user groups, including local residents, wildlife experts, and government officials.

10. Cultural sensitivity testing:
Test the system's ability to be culturally sensitive and responsive to the needs and values of the local community, including testing of outreach and education programs.